

Name OMAATLA BRIDGET MAKWETHE
Course DATA ANALYTICS & DATA SCIENCE

email obmakwethe@gmail.com

Contact: +267 77 052 253

THE LANGUAGE OF BUSINESS ANALYTICS RESEARCH

FOUNDATIONAL CONCEPTS OF ANALYTICS

1. Business Analytics

This involves the use of data, statistical techniques and tools to aid in making informed business decisions. Its primary role is to enhance performance by uncovering trend and insights. In contrast to simple reporting, it prioritizes forecasting and improvement.

Example: A retail business examines sales figures to determine which items in stock to reorder.

2. Data Analytics

Involves the examination of unprocessed (raw) data to identify trends and important information. It is a wide-ranging concept that can be utilized in various sectors and not limited to just business. Business analytics on the other hand, is mostly aimed at informing decisions.

Example: A medical facility reviews patient information to enhance the results of treatments.

3. Business Intelligence

Refers to the methods and technologies employed to gather, display and summarize historical and current data. It emphasizes on reporting and dashboards instead of forecasting.

Example: A business uses Power BI dashboards to monitor sales results on a monthly basis.

4. Data-Driven Decision Making

This involves making choices for a business by using information instead of just gut feelings or predictions. This approach enhances precision and minimizes potential disadvantages.

Examples: A business introduces a new product only after studying data on customer interest.

5. Big Data

Vast information indicates very large/enormous and intricate datasets marked by their size, speed and diversity. Conventional tools struggle to handle it efficiently.

Example: Social networking sites examining millions of user engagements each day.

6. Descriptive Analytics

This entails past events by utilizing historical information. It provides a summary of patterns and trends.

Example: A document illustrating sales results from previous year.

7. Diagnostic Analytics

Clarifies the reasons behind an event by uncovering causes and connections.

Example: Studying the reasons for a decline in sales in a certain area.

8. Predictive Analytics

It employs models along with past data to anticipate future results.

13. Data Warehouse

It is an integrated system designed to keep extensive collections of organized data for examination.

Example: A business maintaining years' worth of sales information for its reports.

14. Data Mining

This involves exploration of trends and connections within extensive data collections.

Examples: Recognizing purchase habits in grocery store sales.

15. Data Visualization

It displays information through various charts and graphs to simplify the understanding of insights.

Example: A bar graph illustrating the increase in revenue.

16. Dashboards

This refers to a control panel that provides an up-to-the-minute visual representation of essential metrics and performance measures.

Example: A marketing control panel showing website visitors and conversion rates.

17. Metric

Refers to any quantifiable figure utilized to monitor performance. Not every metric holds significant importance.

Example: Number of website visitors.

18. Key Performance Indicator (KPI)

It is a particular measurement that clearly shows how well an entity is doing and helps in decision making. Every KPI is a type of metric.

Example: Anticipating customer demand for the upcoming quarter.

9. Prescriptive Analytics

It provides recommendations for actions derived from forecasts and data evaluation.

Example: Advising on the most effective pricing approach to enhance/maximize profitability.

10. Data Literacy

Refers to the skill of interpreting, comprehending and sharing information derived from data. This capability is crucial for making informed decisions today.

Example: A supervisor analyzing dashboard data to inform strategic choices.

DATA, MEASUREMENT AND REPORTING

11. Structured Data

Organized information is arranged in a grid format of rows and columns, which facilitates storage and analysis.

Example: Sales figures in an Excel file.

12. Unorganized Information / Unstructured Data

This is unorganized information that comprises types such as written content, pictures and films that are more challenging to evaluate.

Example: Feedback from customers on a webpage.

Reviews from clients on a website.

but not every metric qualifies as a KPI.

Example: Conversion rate as a KPI for an online store.

19. Benchmark

It is a reference point utilized to evaluate performance in relation to industry standards of rivals.

Example: Comparing company profit margins to industry averages.

20. Segmentation

This means categorizing customers into different groups according to their attributes.

Example: Grouping customers by age or buying behavior.

CUSTOMER AND SALES KPIs

21. Conversion Rate

The proportion of users or clients who accomplish a specific goal.

Example: 5% of website visitors make a purchase.

22. Customer Acquisition Cost (CAC)

The overall expense involved in bringing in a new customer, which covers marketing and sales cost.

Example: Investing P1500.00 to attract one new customer.

23. Customer Lifetime Value (CLV)

The overall profit a company anticipates earning from a customer throughout the duration of their relationship.

Example: A customer generates \$1000 over time.

24. Churn Rate

The portion of clients who discontinue utilizing a product or service.

Example: Each month, 10% of ~~sub~~ subscribers terminate their subscriptions.

25. Customer Retention Rate

The portion of clients that a business maintains over a certain period of time.

Example: Keeping 85% of clients throughout a financial year.

26. Net Promoter Score (NPS)

A metric that assesses customer commitment by their readiness to suggest a product.

Example: Clients rating 9-10 are promoting the product.

27. Average Order Value (AOV)

The typical expenditure for each transaction.

Example: On average, customers spend R50.00 with each purchase they make ~~at~~ at Choppies Tswaneng.

28. Sales Funnel

The process that consumers take from first learning about a product to actually buying it.

Example: Awareness → Interest → Decisions → Purchase.

MARKETING AND CAMPAIGN ANALYTICS

29. Marketing Campaign

A planned series of actions designed to enhance the visibility

of a product or service.

Example: An advertising initiative on social media platforms.

30. Post-Campaign Analysis

This involves assessing how well a marketing campaign performed once it's finished.

Example: Reviewing click-through rates, conversions and revenue generated in an entity.

31. Incremental Revenue

This refers to the extra income that can be directly attributed to a specific marketing campaign.

Example: The additional sales made because of a limited-time discount offer.

32. Return on Investment (ROI)

Evaluates financial gain by examining the relationship between revenue and expenses.

Example: Earning P10,000.00 from a P5,000.00 campaign ($ROI = 100\%$)

33. Return on Ad Spend (ROAS)

The income generated for each pula (currency) invested in advertising/marketing.

Examples: P50 gained for every P10 used on advertising.

34. Click-Through Rate (CTR)

The proportion of individuals who select an advertisement upon viewing it.

Example: 2% of the audience engages with an ad.

35. Bounce Rate

The proportion of users who exit a site without taking any action.

Example: A bounce rate of 70% suggests low user interactions.

36. A/B Testing

Evaluating two variations of content to determine which one is more effective than the other.

Example: Testing two email subject lines.

37. Lead Generation

The method of drawing in and gathering prospective clients through various channels.

Example: Collecting emails through a signup form.

38. Marketing Attribution

This is the process of identifying which marketing channel or touchpoint was responsible for leading a customer to convert.

Example: Identifying whether a social media ad or email drove a sale.

REFERENCES

1. Evans, J. R. (2017). Business Analytics. Pearson
2. McKinsey and Company Insights (mckinsey.com)
3. Statista (statista.com)
4. HubSpot Academy (hubspot.com)
5. Google Analytics Help (support.google.com/analytics)

EXERCISE 2: SQL Fundamentals

SELECTING AND SORTING

1. Retrieve all columns from the employees table.

```
SELECT *
```

```
FROM employees;
```

TABLE employees						
id	first_name	last_name	department	salary	hire-date	city
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

2. Find all unique departments

```
SELECT DISTINCT department
```

```
FROM employees;
```

department
IT
HR
Finance
Marketing

3. Retrieve first and last names ordered by salary descending

```
SELECT first_name, last_name
```

```
FROM employees
```

```
ORDER BY salary DESC;
```

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. Retrieve the top 3 highest-paid employees

```
SELECT id, first-name, last-name, salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;
```

id	first-name	last-name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

Exo

FILTERING WITH WHERE

5. Find employees in the IT department

```
SELECT id, first-name, last-name, department  
FROM employees  
WHERE department = 'Finance' AND salary > 60 000 'IT';
```

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. Find employees in Finance with salary > 60 000

```
SELECT id, first-name, last-name, department, salary  
FROM employees  
WHERE department = 'Finance' AND salary > 60 000;
```

id	first-name	last-name	department	salary
3	Cathy	Zulu	Finance	65 000

7. Find employees in HR or Marketing

```
SELECT id, first_name, last_name, department  
FROM employees  
WHERE department = 'HR' OR 'Marketing';
```

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. Find employees not in IT

```
SELECT id, first_name, last_name, department  
FROM employees  
WHERE department != 'IT';
```

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. Find employees in IT, HR or FINANCE using IN

```
SELECT id, first_name, last_name, department  
FROM employees  
WHERE department  
IN ('IT', 'HR', 'Finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Maidoo	IT

10. Find employees in IT with salary > 65000 and city JHB

```
SELECT id, first_name, last_name, department, salary, city  
FROM employees  
WHERE department = 'IT'  
AND salary > 65000  
AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Maidoo	IT	72 000	Johannesburg

EXERCISE 3: SQL Aggregates and Operators

The students table

Table students

student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

1. List all distinct departments in the student table

```
SELECT DISTINCT department  
FROM students;
```

department
IT
HR
Finance

2. Get the average age of students per department

```
SELECT department, AVG(age) AS average_age
```

```
FROM students
```

```
GROUP BY department;
```

department	avg - age
IT	20.5
HR	22
Finance	23

3. Show departments with more than 1 student

```
SELECT department,
```

```
COUNT (*)
```

```
AS student_count
```

```
GROUP BY department
```

```
HAVING COUNT (*) > 1;
```

department	student_count
IT	2
HR	2

4. Get all students whose age is between 21 and 23-

```
SELECT *
```

```
FROM students
```

```
WHERE age
```

```
BETWEEN 21 AND 23;
```

student id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. List all students in the IT or HR department who are older than 21.

```
SELECT *  
FROM students  
WHERE department IN ('IT', 'HR')  
AND age > 21;
```

student-ID	name	age	department
2	Bob	22	HR
5	Eve	22	HR

The course table

courses			
course ID	course_name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

6. Show total credits per department, only for departments with more than 5 total credits

```
SELECT department, SUM(credits)  
AS total_credits  
FROM courses
```

```
GROUP BY department  
HAVING SUM(credits) > 5;
```

department	total_credits
IT	11

7. List all courses that do not have 4 credits.

SELECT *

FROM courses

WHERE < 4;

course ID	course-name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	IT	3

8. Show the top 3 courses by credits in descending order

SELECT course_ID, course-name, credits

FROM courses

ORDER BY credits DESC

LIMIT 3;

course ID	course name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9. The enrollment table

enrollments			
enroll ID	Student-ID	course ID	Grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

9. Get the maximum, minimum and average grade across all enrollments.

```
SELECT MAX (grade) AS max-grade,  
       MIN (grade) AS min-grade,  
       AVG (grade) AS avg-grade  
FROM   enrollments;
```

max-grade	min-grade	avg-grade
90	78	84.6

10. Count how many enrollments exist per course.

```
SELECT course_id, COUNT (*) AS enrollment-count  
FROM   enrollments  
GROUP BY course-id;
```

course_id	enrollment-count
101	1
102	1
103	1
104	1
105	1

The salaries table

salaries				
employee-id	name	department	salary	bonus
1	Tom	IT	60 000	5000
2	Jerry	HR	55 000	4000
3	Spike	Finance	70 000	6000
4	Tyke	IT	62 000	5500
5	Butch	HR	54 000	3500

11. Find total ^{salary} ~~summary~~ and total bonus per department.

```
SELECT department, SUM(salary) AS total_salary, SUM(bonus)
AS total total - bonus
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

12. Show departments where average salary is above 55 000.

```
SELECT department, AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55 000;
```

department	avg_salary
IT	61 000
Finance	70 000

13. List employees whose salary plus bonus is greater than 60 000.

```
SELECT employee ID, name, salary, bonus, (salary + bonus) AS
total_compensation
```

```
FROM salaries
```

```
WHERE (salary + bonus) > 60 000;
```

employee id	name	salary	bonus	total compensation
1	Tom	60 000	5 000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 000	5 500	67 500

The projects table

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll system	Finance	80 000
3	Dashboard	IT	150 000
4	Website	Marketing	60 000
5	HR Portal	HR	50 000

14. Show total and average budget per department. Only include departments with average budget above 70 000.

```
SELECT departments, SUM(budget) AS total_budget, AVG(budget) AS avg_budget
FROM projects
GROUP BY departments
HAVING AVG(budget) > 70 000;
```

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. List all projects with budgets between 50000 and 120 000 excluding the Marketing department.

```
SELECT * FROM projects
WHERE budget BETWEEN 50000 AND 120 000 AND department <> 'Marketing';
```

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll system	Finance	80 000
5	HR Portal	HR	50 000